



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing–I part–Wednesday, January 10th, 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe the steps followed by a PRAM machine (1)

B. Assuming $P' < P$, is it possible to derive the slowdown factor when P' processors are used? How? (1)

C. Complete the following sentence and explain what it means: "A **write conflict** occurs when..." (1)

D. Amdahl's model is a fixed-size model. True/False? (1)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Briefly describe the difference between Shared address space and Message passing programming models used for parallel programs. (1)

- B. Define what a memory stall is. Explain how stall impact could be mitigated. (1)

- C. Explain when the following sentence is not true: "**A processor with multiple hardware threads has the ability to *avoid stalls*.**" (1)

- D. Write a short description of "**Fine grain multi-threading.**" (1)

Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

- A. On NVIDIA V100 a CUDA grid could be executed on multiple SM units. True/False? (1)

- B. How are scalar registers for CUDA threads organized in an NVIDIA V100SM "sub-core"? (1)

- C. Describe the distinct types of address spaces visible by CUDA kernels. How many? How much is shared? How fast are they? (2)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

A. DRAM Banking could reduce the dead time. True/False? Why? (1)

B. Provide an example of CUDA coalesced accesses. (1)

C. Please explain what a Total Store Ordering consistency model does. (1)

D. In Heterogeneous processing moving less data may give better power consumption. True/False? When? (1)
